

Nguyen Tam Thang

AI Engineer

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ABOUT ME

I am a third-year Computer Science student interested in AI engineering and machine learning. I enjoy building AI systems, experimenting with different approaches, and turning what I learn into practical applications.

EDUCATION

Ho Chi Minh University of Technology (HCMUT)

Bachelor of Computer Science, GPA: 4.0/4.0

Ho Chi Minh City, Vietnam

Sep. 2025 – Nov. 2029 (Expected)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Database Systems, Computer Architecture

PROJECTS

Enterprise Knowledge Intelligence Platform | *Python, FastAPI, Qdrant, Docker, RAG* Aug 2026 – Present

- Built an end-to-end RAG platform covering document ingestion, chunking, BGE embeddings, Qdrant indexing, hybrid Dense + BM25 retrieval, grounded generation, and citations
- Benchmarked Dense, BM25, Weighted RRF, Multi-Query, and reranking strategies on 100 queries; selected Weighted RRF with 0.7247 MRR and 0.9300 Hit@10
- Designed a lightweight cloud retrieval profile using FastEmbed and exact BM25-compatible sparse vectors, achieving 100/100 exact top-10 ranking parity with the local BM25 implementation
- Added API validation, dependency-aware readiness, generation timeouts, admission control, request IDs, Prometheus metrics, and automated CI/regression tests
- Deployed the system publicly with Render, Qdrant Cloud, and Groq using a 130.5 MB Docker image within a 512 MiB memory budget

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java

AI/ML & Retrieval: PyTorch, BGE Embeddings, BM25, Hybrid Search, RRF, SentenceTransformers, FastEmbed

Frameworks & Databases: FastAPI, Qdrant, Pydantic

Tools & Infrastructure: Git, Docker, Docker Compose, GitHub Actions, Prometheus, Grafana, Ollama, Render

CERTIFICATIONS

IELTS Academic: Overall Band 6.5 (CEFR B2)